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Unit – 4 : Session – 10 : SLO – 2 | Heaps

■ I. Fill in the Blanks

- A heap is a special TREE-BASED (complete binary tree) data structure.
- Heaps are usually implemented as ARRAYS.
- A MAX-heap has the largest element at the root.
- A MIN-heap has the smallest element at the root.
- A heap is a COMPLETE binary tree.
- Heaps are used in PRIORITY queues.
- In a max-heap, each parent is GREATER THAN OR EQUAL TO its children.
- In a min-heap, each parent is LESS THAN OR EQUAL TO its children.
- Heap insertion is done at the LAST (next available leaf) position.
- After insertion, we perform HEAPIFY-UP (up-heap / bubble-up) to maintain heap property.
- Deletion is done from the ROOT of the heap.
- After deletion, we perform HEAPIFY-DOWN (down-heap / reheapification).
- Heaps can be efficiently stored using ARRAYS.
- The time complexity for insertion and deletion in a heap is $O(\log n)$.
- Heap sort is a sorting technique based on the HEAP data structure.